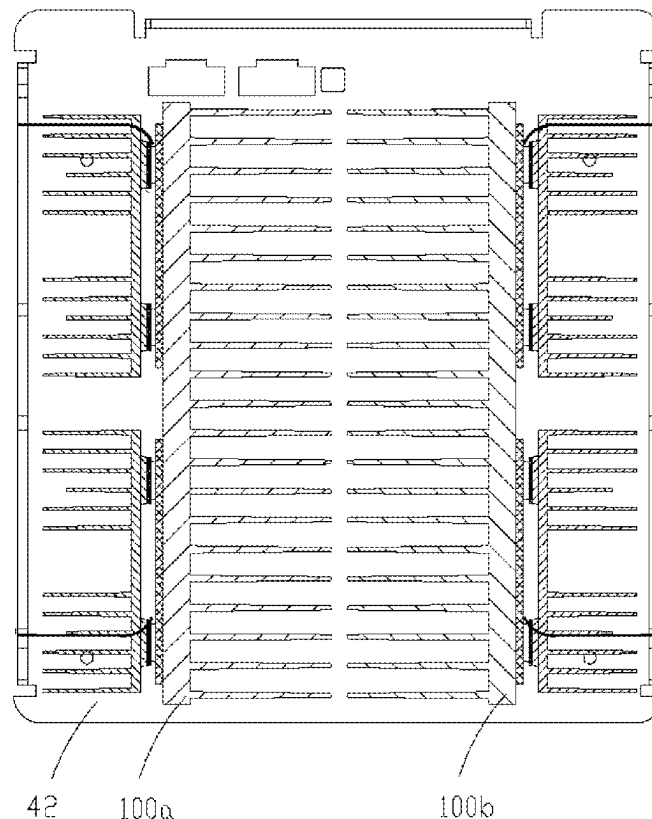


(19) **United States**(12) **Patent Application Publication**
ZHANG et al.(10) **Pub. No.: US 2025/0287526 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **COMPUTATIONAL HEAT DISSIPATION
STRUCTURE, COMPUTING DEVICE
COMPRISING SAME, MINE**(71) Applicant: **Canaan Creative Co., LTD.**, Beijing
(CN)(72) Inventors: **Ning ZHANG**, Beijing (CN); **Nangeng
ZHANG**, Beijing (CN)(21) Appl. No.: **19/218,910**(22) Filed: **May 27, 2025****Related U.S. Application Data**(63) Continuation of application No. 18/544,346, filed on
Dec. 18, 2023, now Pat. No. 12,336,135, which is a
continuation of application No. 18/301,380, filed on
Apr. 17, 2023, now Pat. No. 11,882,669, which is a
continuation of application No. 17/872,172, filed on
Jul. 25, 2022, now Pat. No. 11,895,802, which is a
continuation of application No. 16/614,054, filed on
Nov. 15, 2019, filed as application No. PCT/CN2018/
083956 on Apr. 20, 2018, now Pat. No. 11,425,839.**Foreign Application Priority Data**

May 18, 2017 (CN) 201710353744.5

Publication Classification(51) **Int. Cl.**
H05K 7/20 (2006.01)
F04D 27/00 (2006.01)
G06F 1/20 (2006.01)
(52) **U.S. Cl.**
CPC **H05K 7/20154** (2013.01); **F04D 27/004**
(2013.01); **G06F 1/206** (2013.01); **H05K**
7/20145 (2013.01); **H05K 7/20518** (2013.01)(57) **ABSTRACT**

A computational heat dissipation structure includes a circuit board including a plurality of heating components; and a radiator provided corresponding to the circuit board; wherein a space between the adjacent heating components is negatively correlated with heat dissipation efficiency of a region where the adjacent heating components are located. Since the space between the adjacent heating components of the disclosure is negatively correlated with the heat dissipation efficiency of the region where the adjacent heating components are located, i.e., the higher the heat dissipation efficiency of the region where the adjacent heating components are located is, the smaller the space between the adjacent heating components in the region will be, the heat dissipation efficiencies corresponding to the heating components are balanced, and load of a fan is reduced.



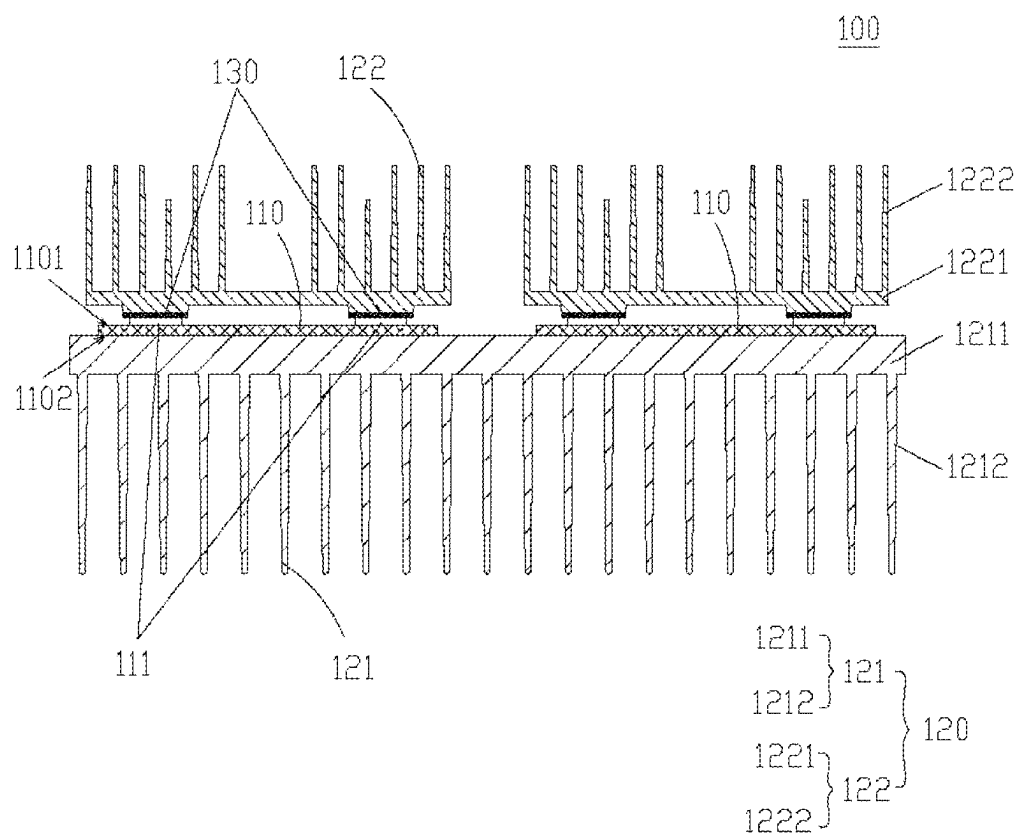


FIG. 1

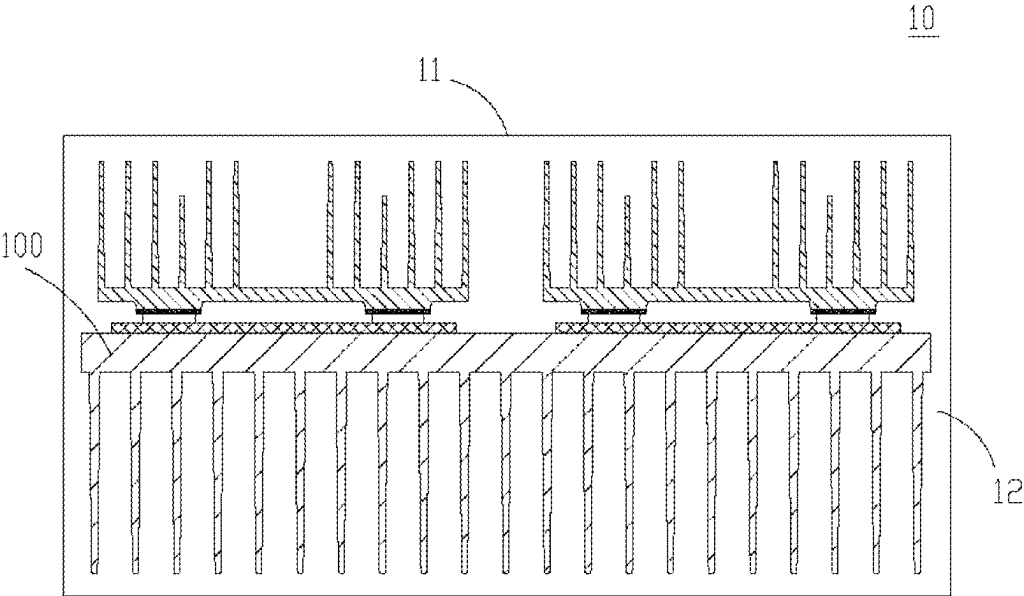


FIG. 2

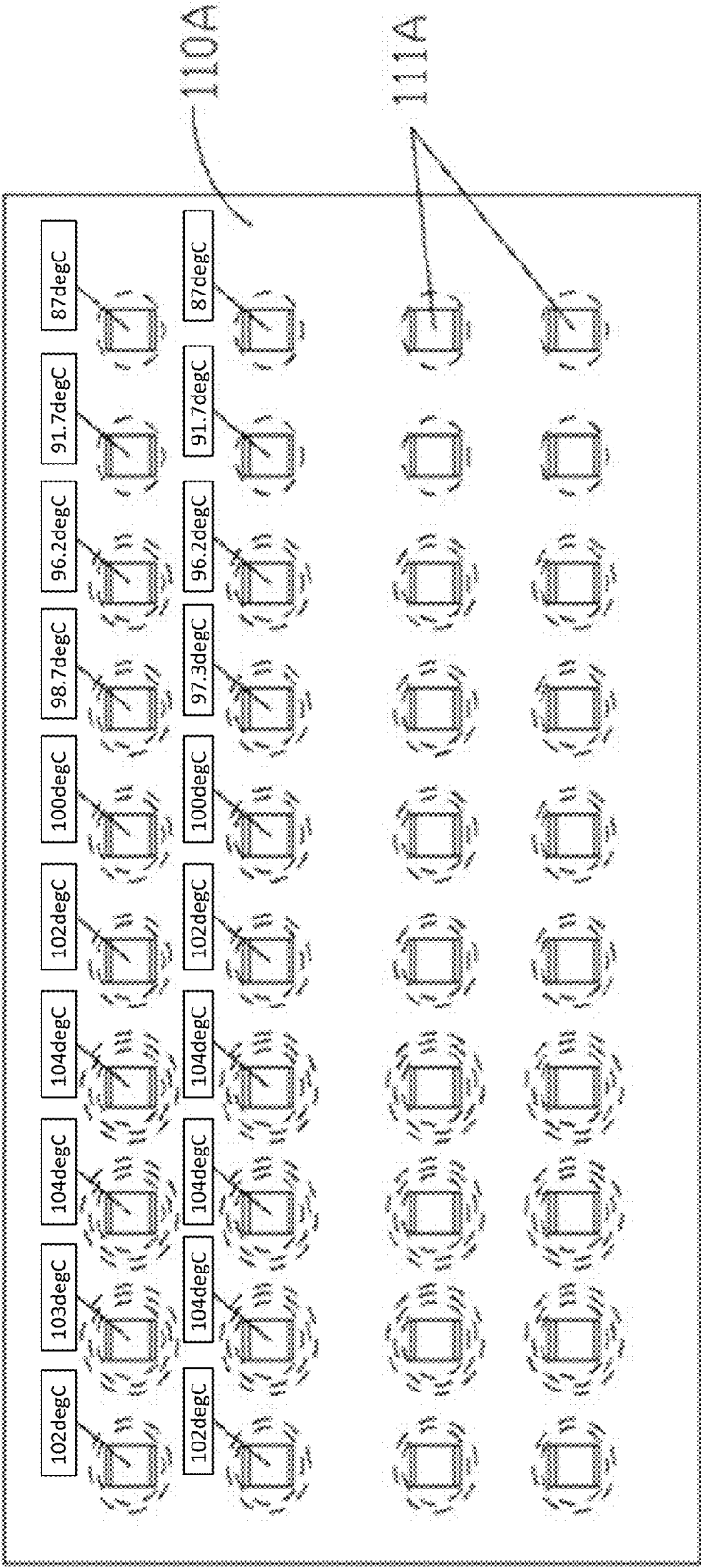


FIG. 3

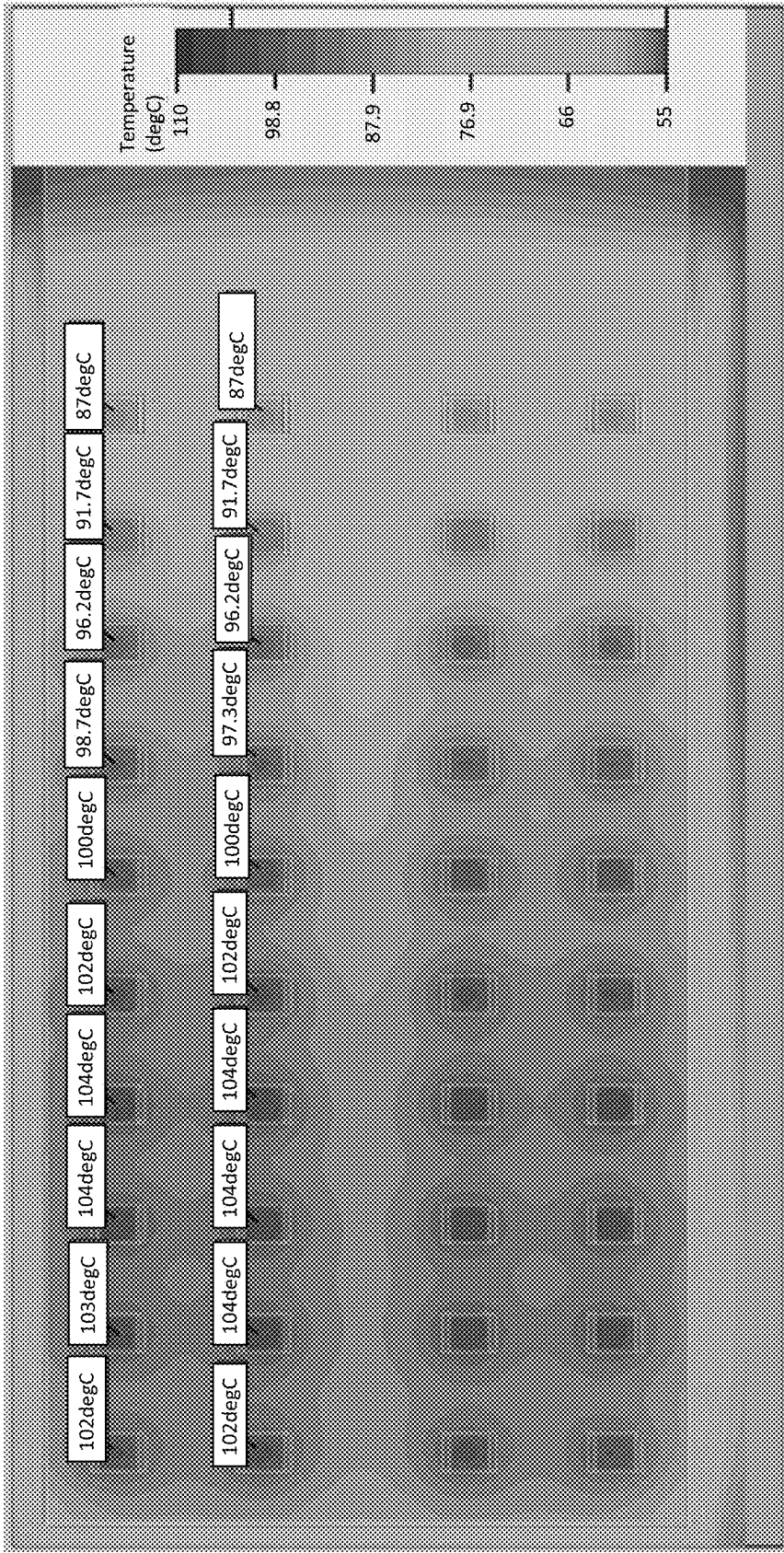


FIG. 4

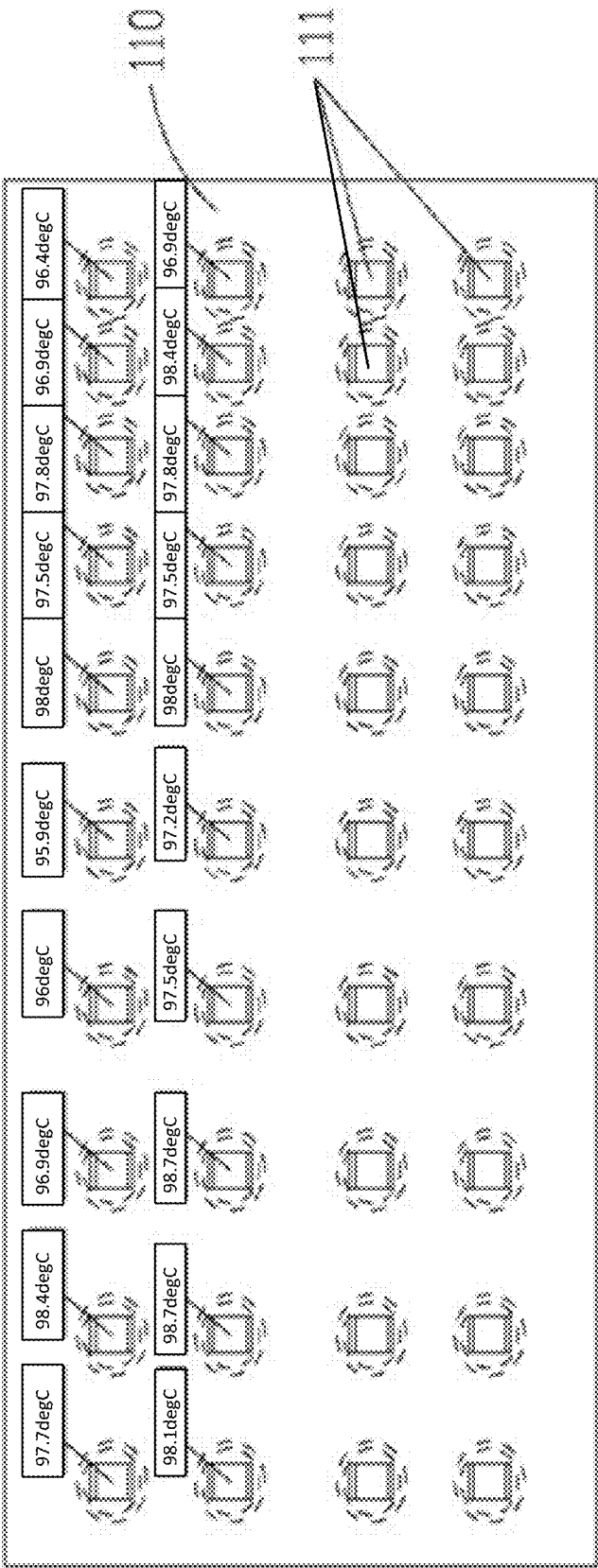


FIG. 5

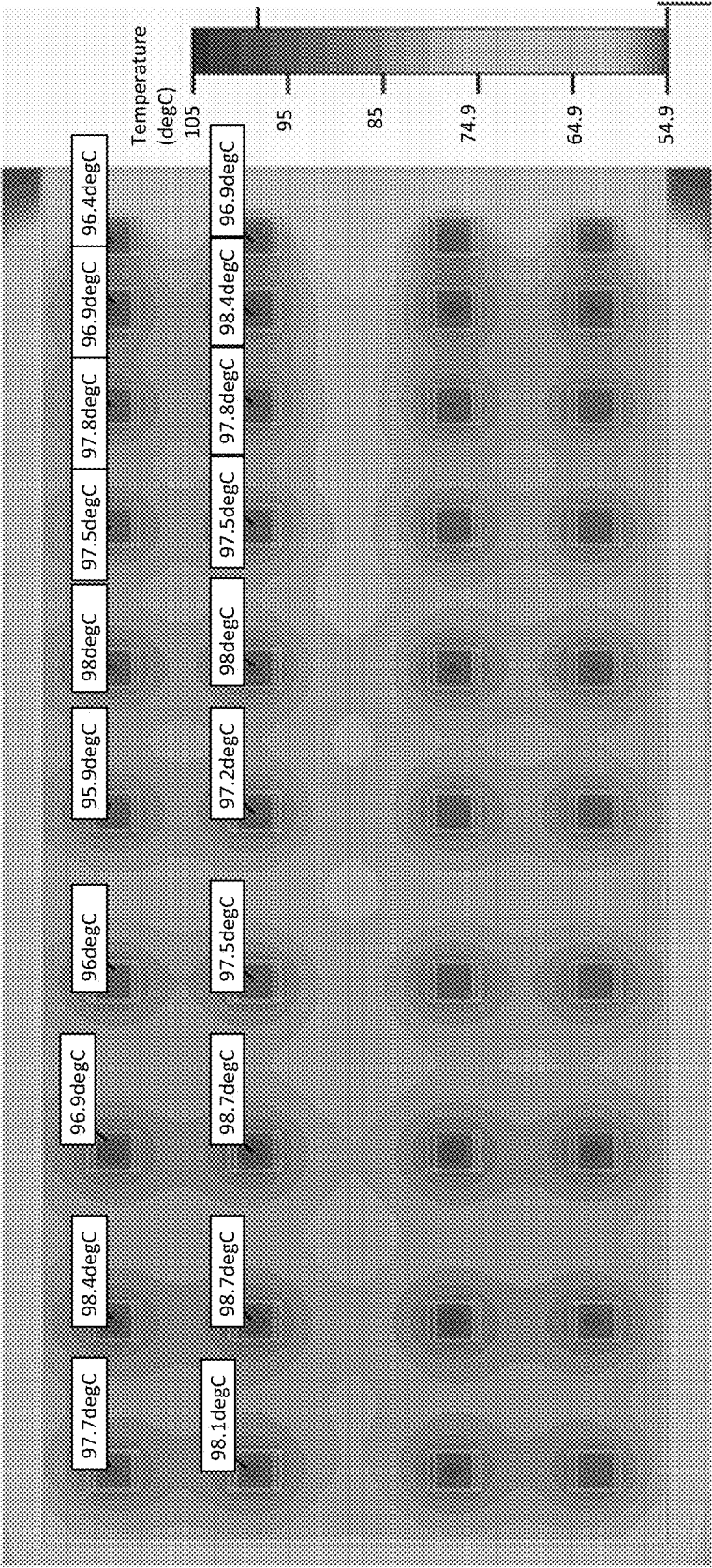


FIG. 6

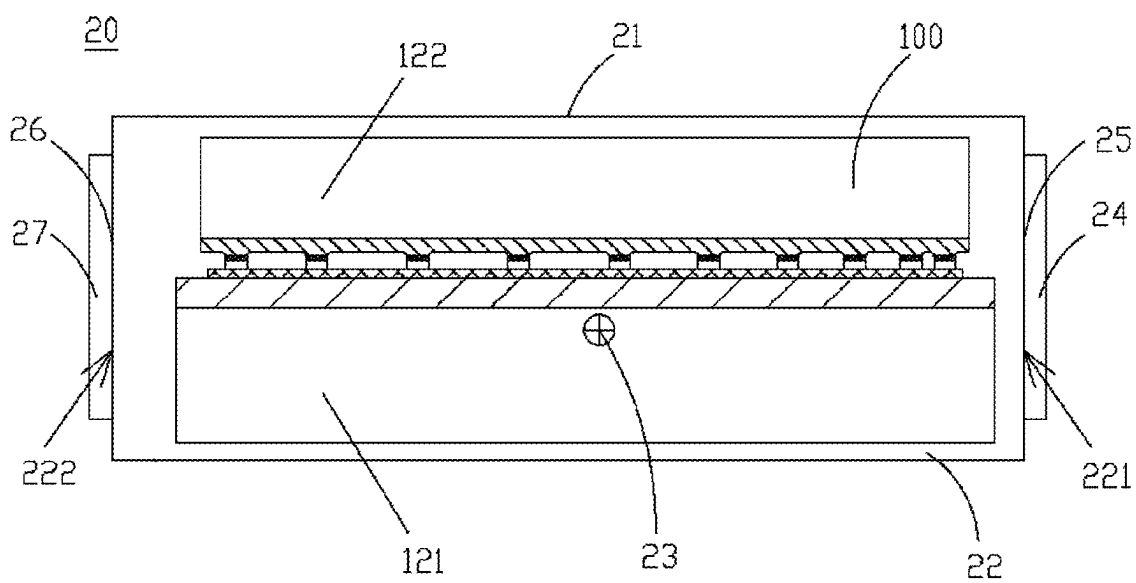


FIG. 7

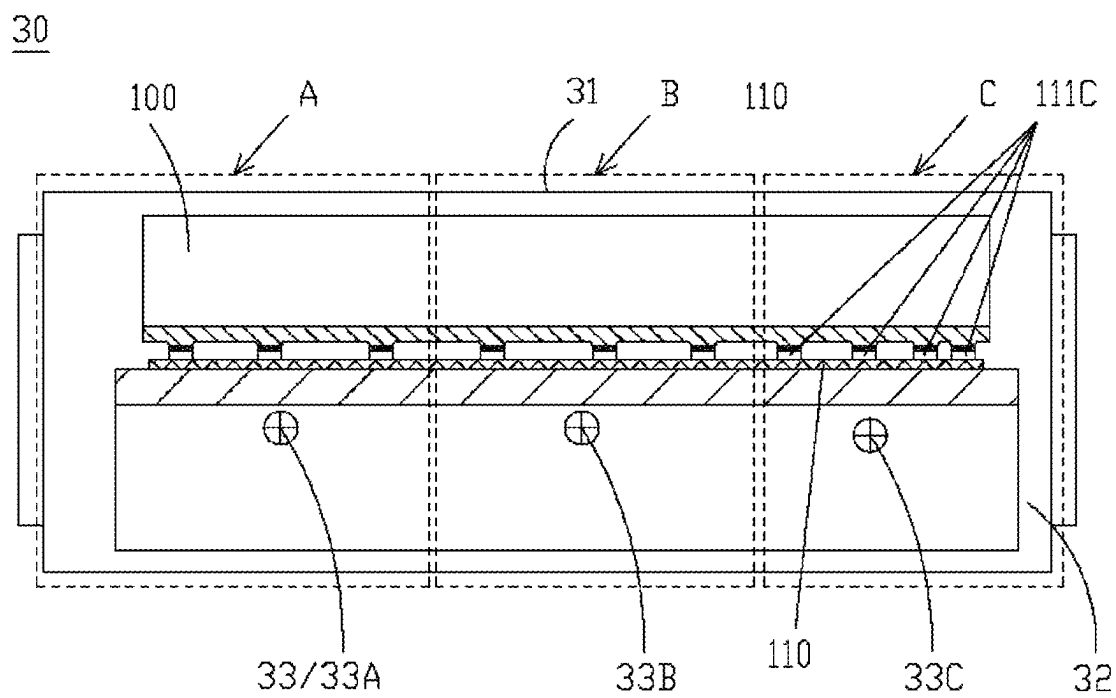


FIG. 8

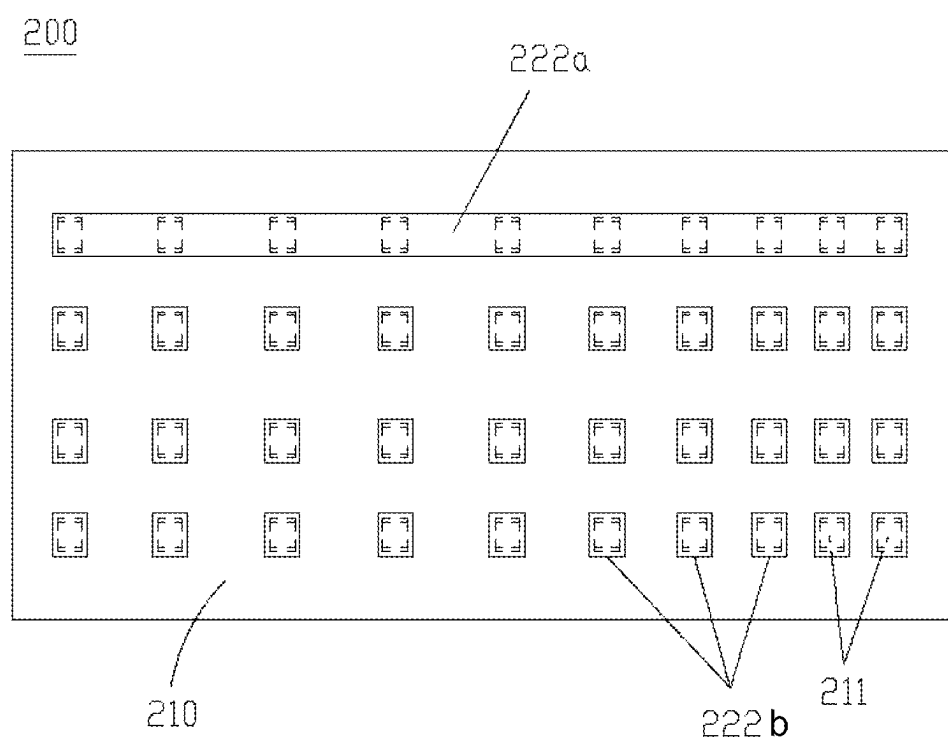


FIG. 9

40

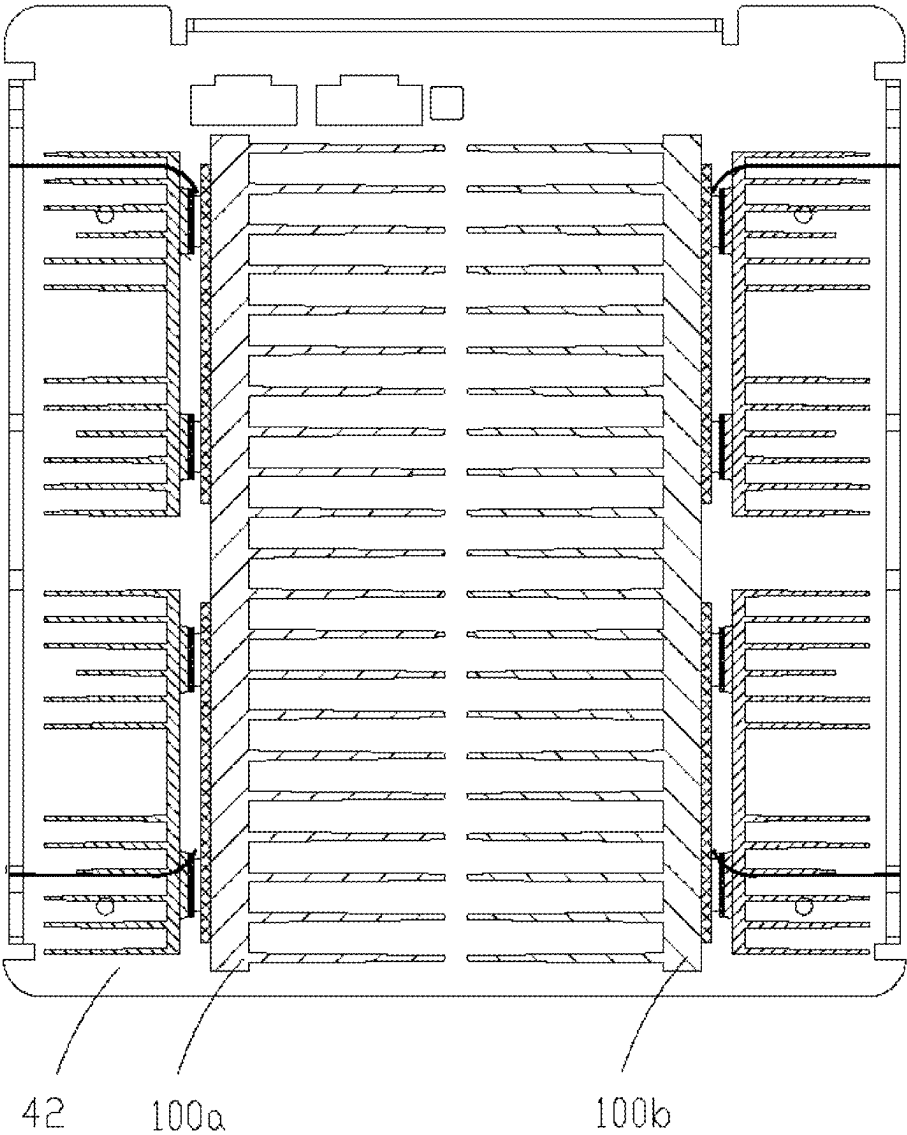


FIG. 10

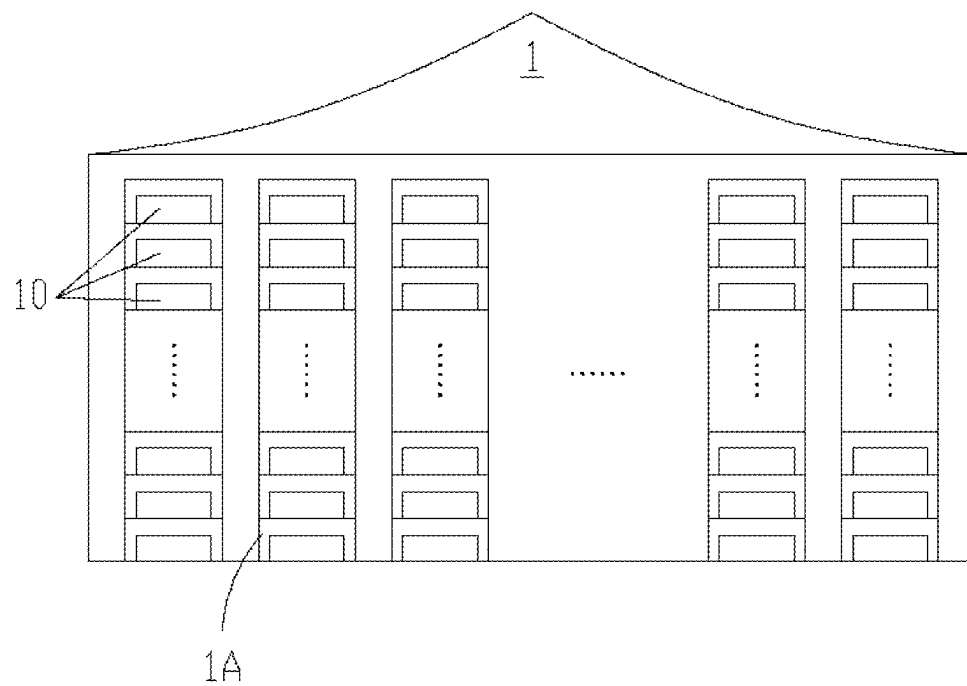


FIG. 11

**COMPUTATIONAL HEAT DISSIPATION
STRUCTURE, COMPUTING DEVICE
COMPRISING SAME, MINE**

**CROSS REFERENCE TO RELATED
APPLICATION**

[0001] This application is a continuation application of U.S. patent application Ser. No. 18/544,346, filed Dec. 18, 2023, which is a continuation application of U.S. patent application Ser. No. 18/301,380, filed on Apr. 17, 2023, which is a continuation application of U.S. patent application Ser. No. 17/872,172, filed on Jul. 25, 2022. U.S. patent application Ser. No. 17/872,172 is a continuation application of U.S. patent application Ser. No. 16/614,054, filed on Nov. 15, 2019. The U.S. patent application Ser. No. 16/614,054 is a national phase application of International Application No. PCT/CN2018/083956, filed with the China National Intellectual Property Administration (CNIPA) on Apr. 20, 2018. International Application No. PCT/CN2018/083956 claims priority to and benefits of Chinese Patent Application No. 201710353744.5, filed with the CNIPA on May 18, 2017. The entire contents of all of the above-identified applications are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The disclosure relates to heat dissipation, and specifically to a computing device and a computational heat dissipation structure thereof. The disclosure further relates to a mine formed of a plurality of the computing devices.

2. Related Art

[0003] In modern computing devices, component integration becomes higher, the number of components deployed on the devices is also increasing, and during operation of the device, the heat generated is also increasing. In a high temperature environment, poor heat dissipation of the components, or uneven heat dissipation of the components at different positions will cause working instability, reduced performance, and shorter working life of the entire circuit. Therefore, the modern computing device has a high requirement for heat dissipation.

[0004] Currently, the heat dissipation method is air cooling. Radiators made of a material with high thermal conductive efficiency are mounted on the high heating components, and then active heat dissipation is carried out through a fan. For example, the Chinese patent for utility model No. ZL201520353408.7 discloses a chip heat dissipation device, a virtual digital coin digger and a computing device, wherein the chip heat dissipation device comprises a chip welded onto a front of a printed circuit board, a first radiator connected to a top of the chip for dissipating heat, and a second radiator connected to a back of the printed circuit board and a part corresponding to a bottom of the chip for dissipating heat, and a fan is located on both sides of a case of the virtual digital coin digger to take away heat generated by the chip in working, and ensure normal working of the digger.

[0005] However, in the prior art, since the chip components are deployed on the circuit board in an equal space, and positions relative to the air duct are different, a temperature difference between an air inlet and an air outlet is

relatively large, and a temperature equalization effect is poor. On such basis, air flowing speed is often increased by increasing rotation speed of the fan, thereby ensuring that the temperature in the hottest region satisfies use requirement, but such method also increases load of the fan.

SUMMARY OF THE INVENTION

[0006] The technical problem to be solved by the disclosure is to provide a computational heat dissipation structure, and a computing device comprising same, and a mine that can balance heat dissipation efficiencies corresponding to respective heating components, and reduce load of a fan.

[0007] To solve the above technical problem, the computational heat dissipation structure of the disclosure comprises a circuit board comprising a plurality of heating components; and a radiator provided corresponding to the circuit board; wherein a space between the adjacent heating components is negatively correlated with heat dissipation efficiency of a region where the adjacent heating components are located.

[0008] In one embodiment of the computational heat dissipation structure, the circuit board comprises a first side face and a second side face, wherein when the circuit board is in a working state, an output heat of the first side face is larger than that of the second side face.

[0009] In one embodiment of the computational heat dissipation structure, the radiator comprises a main heat sink carrying the circuit board; and an auxiliary heat sink pressing the circuit board against the main heat sink tightly.

[0010] In one embodiment of the computational heat dissipation structure, the auxiliary heat sink presses the circuit board against the main heat sink by way of resilient press.

[0011] In one embodiment of the computational heat dissipation structure, a thermal conductive material is filled in a gap between the auxiliary heat sink and the circuit board.

[0012] In one embodiment of the computational heat dissipation structure, the thermal conductive material is thermal conductive silicone grease or thermal conductive rubber pad.

[0013] In one embodiment of the computational heat dissipation structure, heat dissipation efficiency of the main heat sink is higher than that of the auxiliary heat sink, the main heat sink is provided corresponding to the first side face, and the auxiliary heat sink is provided corresponding to the second side face.

[0014] In one embodiment of the computational heat dissipation structure, the main heat sink comprises a first heat dissipation body comprising a first surface and a second surface; and a first heat dissipation fin, wherein the first surface is connected to the first heat dissipation fin, and wherein the second surface is connected to the first side face. The auxiliary heat sink comprises a second heat dissipation body comprising a third surface and a fourth surface; and a second heat dissipation fin, wherein the third surface is connected to the second heat dissipation fin, and wherein the fourth surface is connected to the second side face.

[0015] In one embodiment of the computational heat dissipation structure, there is a plurality of the auxiliary heat sink, each corresponding to one or more of the heating components.

[0016] The computing device of the disclosure comprises a device housing enclosing an enclosed heat-dissipation air duct, and further comprising the above computational heat

dissipation structure, wherein the computational heat dissipation structure is located in the heat-dissipation air duct.

[0017] In one embodiment of the computing device, a temperature detecting unit for measuring a temperature in the heat-dissipation air duct is mounted within the heat-dissipation air duct.

[0018] In one embodiment of the computing device, it further comprises a controller comprising a fan control module; and a fan, the temperature detecting unit and the fan respectively electrically connected to the fan control module; wherein the fan control module adjusts dynamically a rotation speed of the fan based on a real-time temperature in the heat-dissipation air duct detected by the temperature detecting unit.

[0019] In one embodiment of the computing device, the heat-dissipation air duct comprises an air inlet mounted with a first panel assembly; and an air outlet mounted with a second panel assembly; wherein the first panel assembly and the second panel assembly fix the radiator within the heat-dissipation air duct.

[0020] In one embodiment of the computing device, the fan comprises a first fan provided at the air inlet and connected to the first panel assembly.

[0021] In one embodiment of the computing device, the fan comprises a second fan provided at the air outlet and connected to the second panel assembly.

[0022] In one embodiment of the computing device, the heat-dissipation air duct comprises a plurality of temperature zones, and wherein there is a plurality of the temperature detecting unit, each being provided in respective temperature zones.

[0023] In one embodiment of the computing device, the controller further comprises a chip control module electrically connected to respective temperature detecting units, wherein the chip control module adjusts dynamically working frequencies of the heating components in respective temperature zones based on real-time temperature of the temperature zone detected by respective temperature detecting unit.

[0024] In one embodiment of the computing device, there is at least one the computational heat dissipation structure.

[0025] The mine of the disclosure comprises at least one setting frame provided with at least one computing device, wherein the computing device is the above computing device.

[0026] As compared to the prior art, the disclosure has the following advantages: since the space between the adjacent heating components is negatively correlated with the heat dissipation efficiency of the region where the adjacent heating components are located, i.e., the higher the heat dissipation efficiency of the region where the adjacent heating components are located is, the smaller the space between the adjacent heating components in the region will be, the computational heat dissipation structure and the computing device provided by the disclosure balance the heat dissipation efficiencies corresponding to respective heating components, and reduce load of the fan. Therefore, the mine provided by the disclosure avoids the problem of overheating damage of the heating components even if a large number of computing devices are gathered in the same space.

[0027] Further, the disclosure dynamically controls efficiency of the fan and working frequencies of the heating components within different temperature zones, further bal-

ance the heat dissipation efficiencies of different regions in the heat-dissipation air duct, and improve service life of the heating components, entire circuit and fan through temperature data fed back by the temperature detecting units.

[0028] Hereinafter the disclosure is explicitly described with reference to the drawings and the detailed embodiments, but the disclosure is not limited thereto.

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] FIG. 1 is a structure diagram of a computational heat dissipation structure according to one embodiment of the disclosure.

[0030] FIG. 2 is a structure diagram of a computing device according to one embodiment of the disclosure.

[0031] FIG. 3 is a deployment diagram of heating components on a circuit board in the prior art.

[0032] FIG. 4 is a working effect diagram of the heating components on the circuit board of FIG. 3.

[0033] FIG. 5 is a deployment diagram of heating components on a circuit board in the disclosure.

[0034] FIG. 6 is a working effect diagram of the heating components on the circuit board of FIG. 5.

[0035] FIG. 7 is a structure diagram of a computing device according to one embodiment of the disclosure.

[0036] FIG. 8 is a structure diagram of a computing device according to one embodiment of the disclosure.

[0037] FIG. 9 is a structure diagram of a computational heat dissipation structure according to one embodiment of the disclosure.

[0038] FIG. 10 is a structure diagram of a computing device according to one embodiment of the disclosure.

[0039] FIG. 11 is a structure diagram of a mine according to one embodiment of the disclosure.

PREFERABLE EMBODIMENTS OF THE INVENTION

[0040] Hereinafter the technical solution of the disclosure is explicitly described with reference to the drawings and the detailed embodiments to further understand object, solution and effect of the disclosure, instead of limiting the scope protected by the appended claims of the disclosure.

[0041] The disclosure discloses one or more embodiments containing characteristics of the disclosure. The disclosed embodiments are only for exemplary explanation. The scope protected by the disclosure is not limited to the disclosed embodiments. The disclosure is defined by the appended claims.

[0042] References of “one embodiment”, “an embodiment” and “exemplary embodiment” in the specification refer to that the described embodiment can comprise specific features, structures or characteristics, but not every embodiment must comprise these specific features, structures or characteristics. In addition, such expression does not refer to the same embodiment. Further, when the specific feature, structure or characteristic is described with reference to the embodiments, no matter whether it is clearly described, it has shown that combination of such specific feature, structure or characteristic with other embodiments is within the knowledge range of those skilled in the art.

[0043] The specification and subsequent claims use some phrases to refer to specific components or members, those ordinary in the art shall understand that technology users or manufacturers can use different nouns or terms to call the

same component or member. The specification and subsequent claims do not distinguish the components or members by different names, but distinguishing based on functional differences of the components or members as a criterion. “Comprising” and “including” mentioned in the whole specification and subsequent claims are open phrases, and shall be understood as “include but not limited to”. In addition, the word “connection” here includes any direct or indirectly electrical connection means. Indirect electrical connection means comprises connection through other device.

[0044] It shall be noted that terms of “transverse”, “vertical”, “up”, “down”, “front”, “back”, “left”, “right”, “upright”, “horizontal”, “top”, “bottom”, “in”, and “out” indicating orientation or position relations in the disclosure are orientation or position relations based on the drawings in order to facilitate describing the disclosure and simplifying description only, rather than indicating or suggesting that the referred device or element must have a specific orientation, and structured and operated in a specific orientation, so they cannot be understood as limitations to the disclosure.

[0045] FIG. 1 is a structure diagram of a computational heat dissipation structure according to one embodiment of the disclosure. As shown in FIG. 1 and FIG. 5, the computational heat dissipation structure 100 comprises a circuit board 110 comprising a plurality of heating components 111, and a radiator 120, wherein a space between the adjacent heating components 111 is negatively correlated with heat dissipation efficiency of a region where the adjacent heating components 111 are located.

[0046] FIG. 2 is a structure diagram of a computing device according to one embodiment of the disclosure, wherein the computing device comprises the computational heat dissipation structure illustrated in FIG. 1.

[0047] As shown in FIG. 2, the computing device 10 comprises a device housing 11 enclosing an enclosed heat-dissipation air duct 12, and the computational heat dissipation structure 100, wherein the computational heat dissipation structure 100 is located in the heat-dissipation air duct 12.

[0048] FIGS. 3 and 4 are deployment diagrams of heating components on a circuit board in the prior art. As shown in the figures, a left side of the circuit board 110A is close to an air outlet, and a right side is close to an air inlet. As shown in FIG. 3, a plurality of heating components 111A deployed on the device are equally spaced, while a temperature difference between the air outlet and the air inlet is relatively large. Referring to FIGS. 3 and 4, the temperature of the heating component at the air inlet is 87 deg. C., while at a position closer to the air outlet the maximum temperature has reached 104 deg. C. As can be seen, heat dissipation of the heating components 111A is uneven. When temperatures of some heating components 111A are too high due to low heat dissipation efficiency, it might cause working instability, shorter working life and reduced performance of the entire circuit.

[0049] FIGS. 5 and 6 are deployment diagrams of heating components on a circuit board of the computing device in the disclosure. As shown in FIGS. 5 and 6, a left side of the circuit board 110 provided in the disclosure is close to an air outlet, and a right side is close to an air inlet. As shown in the figures, the circuit board 110 is deployed with a plurality of heating components 111, while these heating components 111 are not equally spaced, and a space between the adjacent

heating components is negatively correlated with heat dissipation efficiency of a region where the adjacent heating components are located. Therefore, if the space between the adjacent heating components at the air inlet is small, the space between the adjacent heating components at the air outlet is large, so that temperatures of respective heating components on the circuit board can be balanced.

[0050] As can be seen from the figures, the computational heat dissipation structure deployed with the unequally spaced heating components provided in the disclosure has a maximum temperature of 98.7 deg. C., and a minimum temperature of 95.9 deg. C. with the same rotation speed of the fan as the structure deployed with the equally spaced heating components as shown in FIGS. 3 and 4. As can be seen, with the same rotation speed of the fan, the structure deployed with the unequally spaced heating components provided by the disclosure enables temperatures of the heat dissipation components in respective regions to keep substantially balanced.

[0051] The deployed space of the heating components in respective regions of the circuit board in this embodiment can be obtained through a large number of simulation experiments.

[0052] The advantageous effect of the disclosure lies in: since the space between the adjacent heating components is negatively correlated with the heat dissipation efficiency of the region where the adjacent heating components are located, i.e., the higher the heat dissipation efficiency of the region where the adjacent heating components are located is, the smaller the space between the adjacent heating components in the region will be, the heat dissipation efficiencies corresponding to the heating components are balanced, and load of the fan is reduced.

[0053] In one embodiment, as shown in FIG. 1, the circuit board 110 of the computational heat dissipation structure 100 comprises a first side face 1101 and a second side face 1102.

[0054] In one embodiment, as shown in FIG. 1, the radiator 120 comprises a main heat sink 121 and an auxiliary heat sink 122, wherein the heat dissipation efficiency of the main heat sink 121 is higher than that of the auxiliary heat sink 122.

[0055] In this embodiment, the main heat sink 121 carries the circuit board 110, which can function to dissipate heat, and also function to support the air duct structure, so it functions to dissipate heat and support simultaneously.

[0056] The auxiliary heat sink 122 presses the circuit board 110 against the main heat sink 121, which can function to dissipate heat, and also function to fix the circuit board 110.

[0057] The advantageous effect of this embodiment lies in: since the main heat sink functions to dissipate heat and can also carry the circuit board, it is unnecessary to additionally set other structure for carrying the circuit board, and it saves cost.

[0058] To avoid damage to the heating components 111, in this embodiment, the auxiliary heat sink 122 presses the circuit board 110 against the main heat sink 121 by the way of resilient press. Through such way of press, it can function to fix, and meanwhile, the heating components 111 are not easy to damage since the heating components 111 suffer from a small pressure.

[0059] In one embodiment, as shown in FIG. 1, a thermal conductive material 130 is filled in a gap between the

auxiliary heat sink **122** and the circuit board **110**, and the thermal conductive material **130** is, for example, thermal conductive silicone grease or thermal conductive rubber pad.

[0060] In this embodiment, since the auxiliary heat sink **122** presses the circuit board **110** against the main heat sink **121** by the way of resilient press, a gap exists unavoidably between the auxiliary heat sink **122** and the circuit board **110** to affect heat dissipation effect. Considering of such case, thermal conductive silicone grease or thermal conductive rubber pad is filled in the gap between the auxiliary heat sink **122** and the circuit board **110**. A heat dissipation area of the circuit board **110** is increased in a disguised form by filling thermal conductive silicone grease or thermal conductive rubber pad in the gap between the auxiliary heat sink **122** and the circuit board **110**, thereby further enhancing the heat dissipation effect of the circuit board **110**.

[0061] The advantageous effect of this embodiment lies in that the heat dissipation effect of the circuit board is further enhanced by filling the thermal conductive material in the gap between the auxiliary heat sink and the circuit board.

[0062] In one embodiment, as shown in FIG. 1, since the heat dissipation efficiency of the main heat sink **121** is higher than that of the auxiliary heat sink **122**, the main heat sink **121** can be provided corresponding to the second side face **1102** of the circuit board **110**, and the auxiliary heat sink **122** can be provided corresponding to the first side face **1101** of the circuit board **110**.

[0063] Specifically, the main heat sink **121** comprises a first heat dissipation body **1211** comprising a first surface and a second surface opposite to each other, and a first heat dissipation fin **1212** connected to the first surface of the first heat dissipation body **1211**, and the second surface of the first heat dissipation body **1211** connected to the second side face **1102** of the circuit board **110**. The auxiliary heat sink **122** comprises a second heat dissipation body **1221** comprising a third surface and a fourth surface opposite to each other, and a second heat dissipation fin **1222** connected to the third surface of the second heat dissipation body **1221**, and the fourth surface of the second heat dissipation body **1221** connected to the first side face **1101** of the circuit board **110**.

[0064] Because of the main heat sink and the auxiliary heat sink, heat on two sides of the circuit board is balanced, thereby further balancing heat of respective regions in the heat-dissipation air duct.

[0065] In one embodiment, as shown in FIG. 7, a computing device **20** comprises a device housing **21** enclosing an enclosed heat-dissipation air duct **22**, and the computational heat dissipation structure **100**, wherein the computational heat dissipation structure **100** is located in the heat-dissipation air duct **22**. The heat-dissipation air duct **22** is mounted with a temperature detecting unit **23** for measuring a temperature in the heat-dissipation air duct **22**.

[0066] In one embodiment, as shown in FIG. 8, a computing device **30** comprises a device housing **31** enclosing an enclosed heat-dissipation air duct **32**, and the computational heat dissipation structure **100**, wherein the computational heat dissipation structure **100** is located in the heat-dissipation air duct **32**. The heat-dissipation air duct **32** is mounted with a plurality of temperature detecting units **33** for measuring temperatures in the heat-dissipation air duct **32**, and the plurality of temperature detecting units **33** are

provided in different regions of the heat-dissipation air duct **32** to real-time detect temperatures of different regions.

[0067] In this embodiment, the temperature detecting units are mounted in the heat-dissipation air duct to record temperatures in the heat-dissipation air duct, so that during working of the computing device, the user can acquire the temperatures in the heat-dissipation air duct, and adjusts a rotation speed of the fan.

[0068] In one embodiment, as shown in FIG. 7, the computing device **20** further comprises a controller comprising a fan control module and comprises a fan **24**, the temperature detecting unit **23** and the fan **24** respectively electrically connected to the fan control module. The fan control module dynamically adjusts a rotation speed of the fan **24** based on the temperature in the heat-dissipation air duct **22** detected by the temperature detecting unit **23**, so that the fan **24** is avoided to keep a high rotation speed for a long time, thereby reducing electric energy loss.

[0069] In this embodiment, the rotation speed of the fan is automatically adjusted by the fan control module, or manually adjusted by giving an instruction based on the temperature fed back from the temperature detecting unit.

[0070] In one embodiment, as shown in FIG. 7, the heat-dissipation air duct **22** comprises an air inlet **221** mounted with a first panel assembly **25**, and an air outlet **222** mounted with a second panel assembly **26**. The first panel assembly **25** and the second panel assembly **26** fix the radiator **120**, i.e., the main heat sink **121** and the auxiliary heat sink **122**, within the heat-dissipation air duct **22** in parallel.

[0071] The fan **24** is, for example, fixed onto the first panel assembly **25** at the air inlet **221**.

[0072] In one embodiment, as shown in FIG. 7, the computing device **20** further comprises a fan **27** provided at the air outlet **222**, and connected to the second panel assembly **26**.

[0073] In this embodiment, the computing device **20** is provided with the fan **24** and the fan **27** on both ends. The fan **24** at the air inlet **221** blows air, and the fan **27** at the air outlet absorbs air, so that a faster and more fluent heat dissipation airflow is formed in the heat-dissipation air duct **22**, and the heat dissipation effect is further enhanced.

[0074] According to the respective embodiments, it shall be noted that the position and the number of fan in the disclosure can be adjusted according to the need. For example, the fan **27** can be provided only at the air outlet **222** to suction out air, the fan **24** can be provided only at the air inlet **221** to blow air, and the fans also can be provided at both the air outlet **222** and the air inlet **221** to enhance airflow, but the disclosure is not limited thereto.

[0075] In one embodiment, as shown in FIG. 8, the heat-dissipation air duct **32** is divided into, for example, three temperature zones, which are a first temperature zone A, a second temperature zone B, and a third temperature zone C, respectively. The temperature detecting units are, for example, three, which are the temperature detecting units **33A**, **33B** and **33C**, respectively, and the temperature detecting units **33A**, **33B** and **33C** respectively detect temperatures of the first temperature zone A, the second temperature zone B, and the third temperature zone C of the heat-dissipation air duct **32**. Of course, each temperature zone can also correspond to more than one temperature detecting unit.

[0076] The first temperature zone A, the second temperature zone B, and the third temperature zone C correspond to a part of the heating components on the circuit board 110, respectively.

[0077] The controller of the computing device 30 further comprises a chip control module electrically connected to the temperature detecting units 33A, 33B and 33C, wherein the chip control module adjusts dynamically working frequencies of the heating components in the temperature zones where the temperature detecting units 33A, 33B and 33C are located based on the temperatures of different temperature zones A, B and C in the heat-dissipation air duct 32 detected by the temperature detecting units 33A, 33B and 33C.

[0078] For example, when the temperature of the third temperature zone C detected by the temperature detecting unit 33C is too high, the chip control module controls a part of the heating components 111C corresponding to the third temperature zone C to reduce the working frequencies, so as to decrease the heat of this part of the heating components 111C, and balance the heat dissipation efficiencies of respective heating components.

[0079] In this embodiment, life of the heating components can be prolonged, and problems of too much noise due to long-time overload working of the fan, and overall working efficiency affected by damage of the fan and the circuit can be avoided by reducing the working efficiencies of a part of the heating components in an overheating region.

[0080] FIG. 9 is a structure diagram of a computational heat dissipation structure according to one embodiment of the disclosure. In one embodiment, as shown in FIG. 9, the computational heat dissipation structure 200 comprises a circuit board 210 and a radiator. The circuit board 210 is provided with a plurality of heating components 211, which are unequally spaced, and a space between the adjacent heating components 211 is negatively correlated with heat dissipation efficiency of a region where the adjacent heating components 211 are located.

[0081] The radiator comprises a main heat sink (not shown) and an auxiliary heat sink provided on two side faces of the circuit board 210 respectively. For example, there may be a plurality of auxiliary heat sinks 222b, each being provided corresponding to one of the heating components 211. Of course, each auxiliary heat sink 222a may also be provided corresponding to more of the heating components 211. It shall be noted that in different embodiments, the number of circuit board 210 may be one or more, but the disclosure is not limited thereto.

[0082] According to the disclosure, there may be only one computational heat dissipation structure in the heat-dissipation air duct as the computing device shown in FIGS. 1, 7 and 8, and alternatively there may be two computational heat dissipation structures 100a and 110b in a heat-dissipation air duct 42 as a computing device 40 shown in FIG. 10, wherein the computational heat dissipation structures 100a and 110b are symmetrically provided. Of course, more computational heat dissipation structures can also be included.

[0083] The computing device of the disclosure can be a computing device for a virtual coin mining machine, and a mine in the disclosure refers to a place where a large number of mining machines are gathered for virtual coin mining.

[0084] The structure of the mine in one embodiment of the disclosure is shown in FIG. 11. The mine 1 comprises at least one setting frame 1A provided with at least one computing device of the disclosure, for example, the com-

puting device 10. It shall be noted that the setting frame 1A can have computing devices in various ways, for example, parallel and stack, and the structure and the form of the setting frame 1A are not limited to the figures.

[0085] Of course, the disclosure also can have other various embodiments, and those skilled in the art can make various corresponding modifications and variations based on the disclosure without departing from spirit and essence of the disclosure, but these corresponding modifications and variations shall belong to the scope protected by the appended claims of the disclosure.

INDUSTRIAL APPLICABILITY

[0086] Since the space between the adjacent heating components is negatively correlated with the heat dissipation efficiency of the region where the adjacent heating components are located, i.e., the higher the heat dissipation efficiency of the region where the adjacent heating components are located is, the smaller the space between the adjacent heating components in the region will be, the computational heat dissipation structure and the computing device provided by the disclosure balance the heat dissipation efficiencies of respective heating components, and reduce load of the fan. Therefore, the mine provided by the disclosure avoids the problem of overheating damage of the heating components even if a large number of computing devices are gathered in the same space.

What is claimed is:

1. A circuit board, comprising:

a heat-generating component array is provided on a surface of the circuit board,

wherein the heat-generating component array comprises a plurality of heat-generating components of a same specification arranged in rows and columns,

at least two sub-arrays are arranged in parallel in a column direction of the heat-generating component array, and a spacing in the column direction between two adjacent rows of heat-generating components in a same sub-array of the at least two sub-arrays is not equal to a spacing in the column direction between two adjacent sub-arrays of the at least two sub-arrays.

2. The circuit board of claim 1, wherein:

the spacing in the column direction between two adjacent rows of heat-generating components in the same sub-array is smaller than the spacing in the column direction between the two adjacent sub-arrays.

3. The circuit board of claim 1, wherein:

the spacing in the column direction between the two adjacent sub-arrays is smaller than a size of each sub-array of the at least two sub-arrays in the column direction.

4. The circuit board of claim 1, wherein:

a size of each sub-array of the at least two sub-arrays in a row direction of the heat-generating component array is greater than a size of the sub-array in the column direction.

5. The circuit board of claim 4, wherein:

the size of each sub-array in the row direction of the heat-generating component array is more than five times the size of the sub-array in the column direction.

6. The circuit board of claim 1, wherein:

a first side edge of the at least two sub-arrays in the row direction is parallel to the column direction; and/or

a second side edge of the at least two sub-arrays in the row direction is parallel to the row direction.

7. The circuit board of claim 1, wherein:

in the column direction, two adjacent sub-arrays of the at least two sub-arrays are symmetrically arranged based on a central axis between the two adjacent sub-arrays.

8. The circuit board of claim 1, wherein:

in each sub-array, a spacing between adjacent heat-generating components in the same row comprises at least three lengths,

wherein, among the at least three lengths, the spacing with a smallest length is distributed closer to a front along a heat dissipation direction relative to the spacing with a largest length.

9. The circuit board of claim 1, characterized in that each sub-array of the at least two sub-arrays includes a same number of heat-generating components.

10. The circuit board of claim 1, characterized in that each sub-array of the at least two sub-arrays comprises at least two rows of heat-generating components.

11. The circuit board of claim 1, characterized in that each sub-array of the at least two sub-arrays comprises at least ten columns of heat-generating components.

12. The circuit board of claim 10, characterized in that each row in each sub-array comprises a same number of heat-generating components.

13. The circuit board of claim 1, wherein:

along a row direction of the heat-generating component array, an average spacing between the foremost five heat-generating components in a same row in each sub-array is smaller than an average spacing between the rearmost five heat-generating components in the same row.

14. The circuit board of claim 1, wherein:

when the circuit board is placed in an air duct having an air inlet and an air outlet, a direction from the air inlet to the air outlet is a heat dissipation direction.

15. The circuit board of claim 14, wherein:

for a same row of heat-generating components in each sub-array, a spacing of a first pair of adjacent heat-generating components near the air inlet is smaller than a spacing of a first pair of adjacent heat-generating components near the air outlet.

16. The circuit board of claim 14, wherein:

in a same row of heat-generating components in each sub-array, a spacing between at least a subset of adjacent heat-generating components is positively correlated with a distance from the subset of adjacent heat-generating components to the air inlet; and/or

a spacing length between at least a subset of adjacent heat-generating components is negatively correlated with a distance from the subset of adjacent heat-generating components to the air outlet.

17. The circuit board of claim 1, wherein:

a spacing between adjacent heat-generating components in the same row in each sub-array is negatively correlated with a heat dissipation efficiency of a region where the adjacent heat-generating components are located.

18. The circuit board of claim 1, characterized in that the plurality of heat-generating components of the same specification comprise:

a plurality of heat-generating components having a same size, and/or

a plurality of heat-generating components having a same heating area, and/or

a plurality of heat-generating components having a same model.

19. A working assembly, comprising the circuit board of claim 1 and a radiator module provided on the surface of the circuit board.

20. A computing device, comprising a power supply and the working assembly of claim 19.

* * * * *